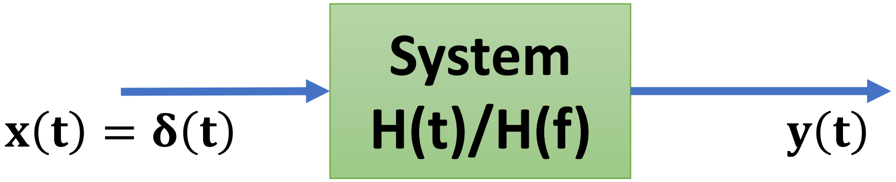
Convolution

Outlines

- Basics of Convolution
- Importance of Convolution
- Function and Formulas of Convolution
- Properties of Convolution

Basics of Convolution

- The Impulse Response of the LTI System can be Identified by Convolution.



- Output of the System:

$$\Rightarrow y(t) = x(t) * h(t)$$

$$\Rightarrow y(f) = x(f) \times h(f)$$

- Using Convolution, we can identify the Zero-State Response of the LTI System.
- Convolution is used in the following fields:
  - Signal and Image Processing
  - Control Systems
  - Pattern Recognition
  - Neural Networks

Function and Formulas of Convolution

- Convolution of Continuous Time Signals $f(t)$ and $g(t)$ is given by:

$$\Rightarrow f(t) * g(t) = \int_{-\infty}^{\infty} f(\tau)g(t-\tau)d\tau = \int_{-\infty}^{\infty} f(t-\tau)g(\tau)d\tau$$

- Convolution of Discrete Time Signals $f(n)$ and $g(n)$ is given by:

$$\Rightarrow f(n) * g(n) = \sum_{k=-\infty}^{\infty} f(k)g(n-k) = \sum_{k=-\infty}^{\infty} f(n-k)g(k)$$

- Convolution in the time domain is Multiplication in the frequency domain.

Properties of Convolution

- **Commutative Property:**

$$\Rightarrow f(t) * g(t) = g(t) * f(t)$$

- **Distributive Property:**

$$\Rightarrow f(t) * \big(g_1(t) + g_2(t)\big) = f(t) * g_1(t) + f(t) * g_2(t)$$

- **Associative Property:**

$$\Rightarrow f(t) * \big(g(t) * h(t)\big) = \big(f(t) * g(t)\big) * h(t)$$

- **Time Shifting Property:**

$$\Rightarrow f(t) * g(t) = h(t)$$

$$\Rightarrow f(t) * g(t - t_0) = h(t - t_0)$$

- **Example 1**: If $x(t) * y(t) = z(t)$ then $x(t-3) * y(t+1) = z(?)$
- Solution: $x(t-3) * y(t+1) = z(t-2)$

- **Example 2**: If $x(t) * y(t) = z(t)$ then $x(t+3) * y(t+1) = z(?)$
- Solution: $x(t+3) * y(t+1) = z(t+4)$

Properties of Convolution

❑ Convolution with Impulse Signal:

$$\Rightarrow f(t) * \delta(t) = f(t)$$

$$\Rightarrow f(t) * \delta(t - t_0) = f(t - t_0)$$

❑ Essential Property Impulse Signal:

$$\Rightarrow \delta(at) = \left(\frac{1}{|a|}\right)\delta(t)$$

❑ Differentiation Property:

$$\Rightarrow f(t) * g(t) = h(t)$$

$$\Rightarrow f(t) * \frac{dg(t)}{dt} = \frac{df(t)}{dt} * g(t) = \frac{dh(t)}{dt}$$

$$\Rightarrow \frac{d^m f(t)}{dt} * \frac{d^n g(t)}{dt} = \frac{d^{m+n} h(t)}{dt}$$

❑ Time Scaling Property:

$$\Rightarrow f(t) * g(t) = h(t)$$

$$\Rightarrow f(at) * g(at) = \left(\frac{1}{|a|}\right)h(at)$$

- **Example 1**: $x(t-3) * \delta(2t+4) = x(?)$
- **Solution:**

$$= x(t-3) * \delta(2t+4)$$

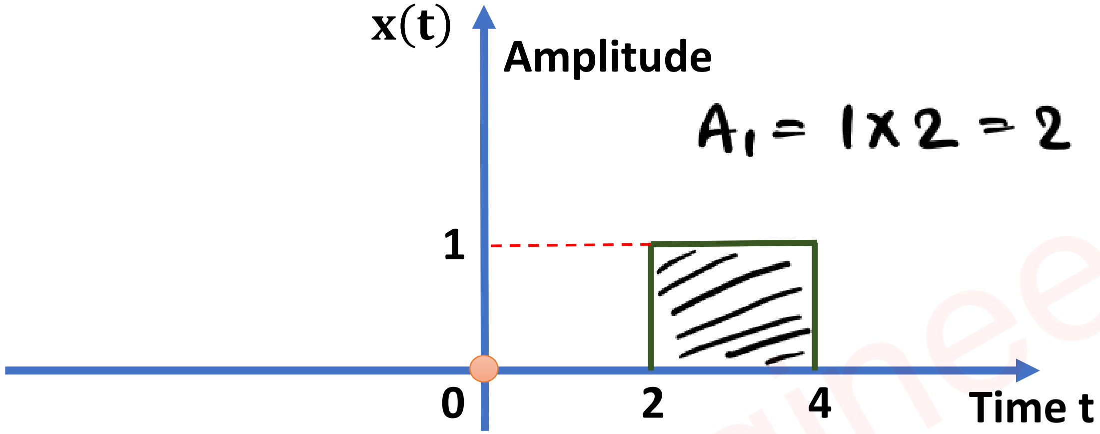
$$= x(t-3) * \delta(2(t+2))$$

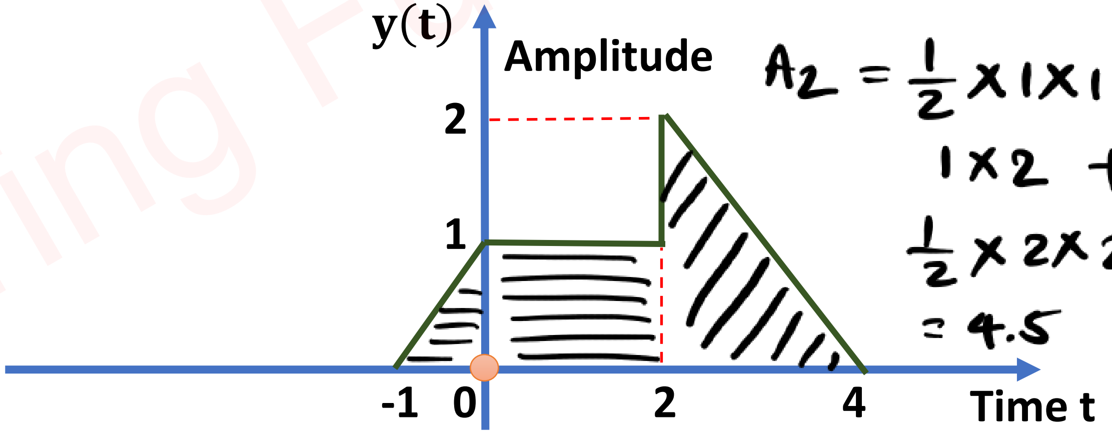
$$= x(t-3) * \left(\frac{1}{2}\right)\delta((t+2))$$

$$= \left(\frac{1}{2}\right)x((t-1))$$

Limits of Convolution and Area of Convoluted Signal

Example 1. Identify the range of convoluted signal $z(t)$ and also, identify area of convoluted Signal with respect to time axis. Here, $z(t) = x(t) * y(t)$





$A_2 = \frac{1}{2} \times 1 \times 1 + 1 \times 2 + \frac{1}{2} \times 2 \times 2 = 4.5$

→ lower limits $= 2 + (-1) = 1$

Upper limits $= 4 + (4) = 8$

→ Range of Convoluted Signal $z(t)$.

$\Rightarrow \underline{1 < t < 8}$

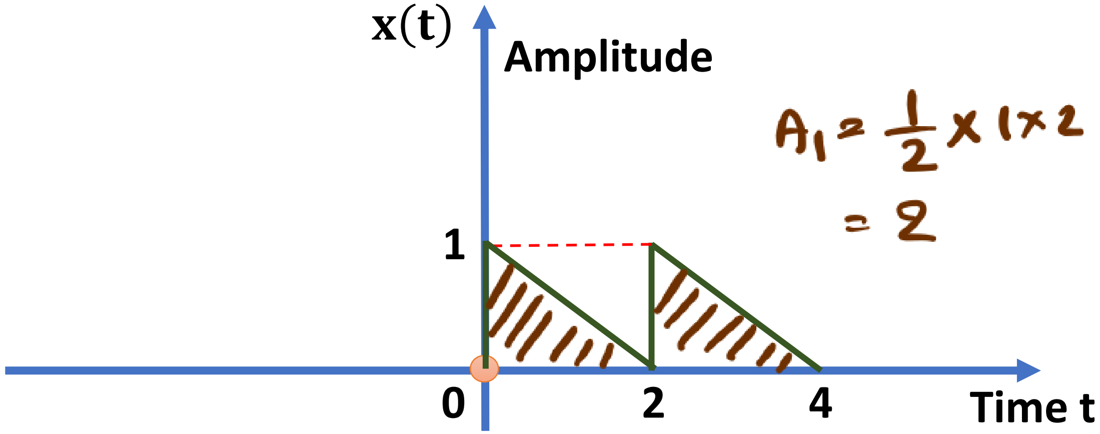
→ Area of Convoluted Signal $z(t)$

$= A_1 A_2$

$= 2 \times 4.5$

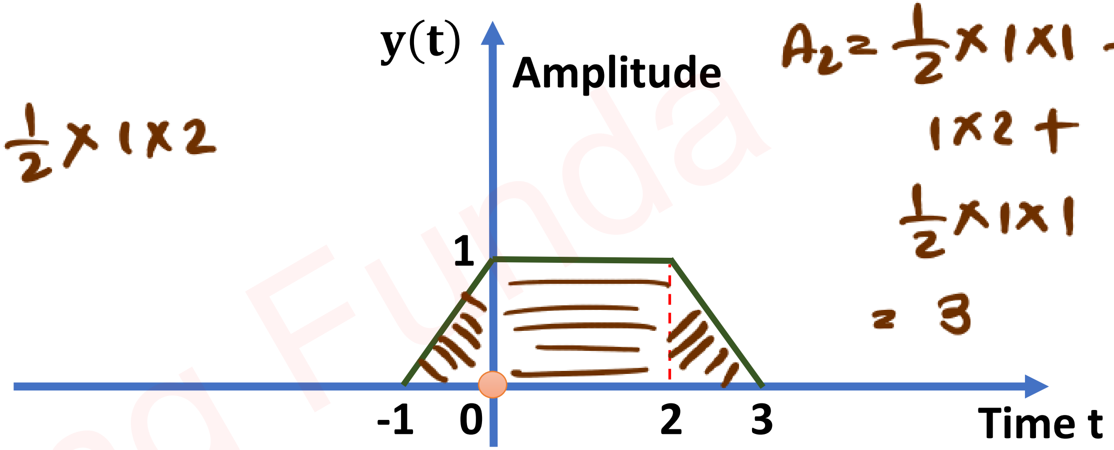
$= 9$

Example 2. Identify the range of convoluted signal $z(t)$ and also, identify area of convoluted Signal with respect to time axis. Here, $z(t) = x(t) * y(t)$



$A_1 = \frac{1}{2} \times 1 \times 2 + \frac{1}{2} \times 1 \times 2$

$= 2$



$A_2 = \frac{1}{2} \times 1 \times 1 + 1 \times 2 + \frac{1}{2} \times 1 \times 1$

$= 3$

~ lower limits $= 0 + (-1) = -1$

Upper limits $= 4 + (3) = 7$

~ Range of Convoluted signal $z(t)$

$\Rightarrow \quad -1 < t < 7$

~ Area of Convoluted Signal $z(t)$

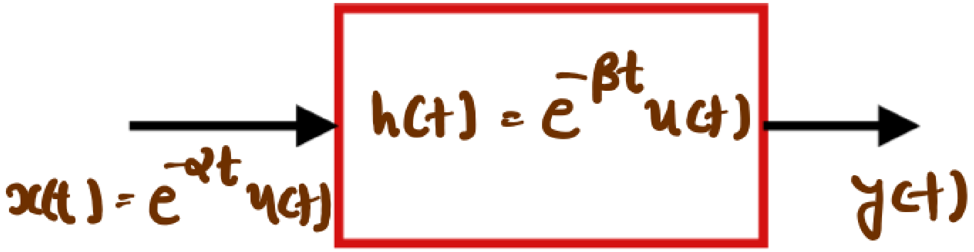
$= A_1 A_2$

$= 2 \times 3$

$= 6$

Convolution of Continuous Time Signals

Example 1. Using the convolution integral, find the response $y(t)$ of a LTI system with impulse response $h(t) = e^{-\beta t}u(t)$ to the input $x(t) = e^{-\alpha t}u(t)$.

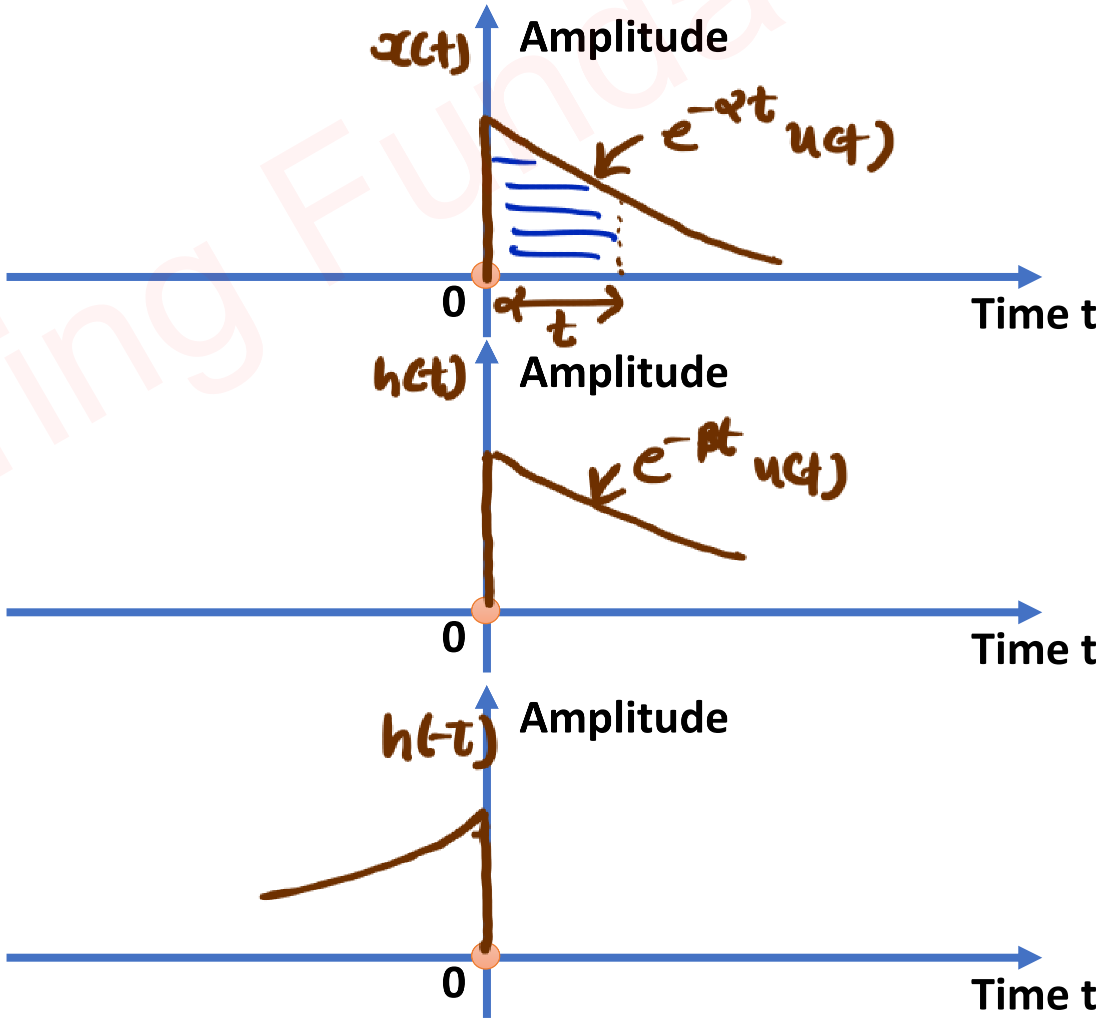


$$y(t) = x(t) * h(t)$$

$$= \int_{-\infty}^{\infty} x(\tau)\, h(t-\tau)\, d\tau$$

$$= \int_{0}^{t} e^{-\alpha\tau}\, e^{-\beta(t-\tau)}\, d\tau$$

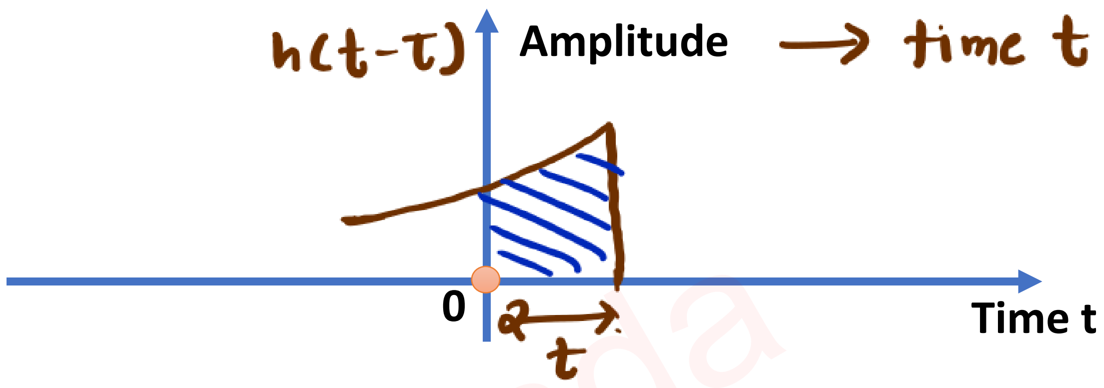
$$= e^{-\beta t} \int_{0}^{t} e^{(\beta-\alpha)\tau}\, d\tau$$



$$= e^{-\beta t}\left[\frac{e^{(\beta-\alpha)\tau}}{\beta-\alpha}\right]_0^t$$

$$= \frac{e^{-\beta t}}{\beta-\alpha}\left[e^{(\beta-\alpha)t} - 1\right]$$

$$y(t) = \frac{e^{-\alpha t} - e^{-\beta t}}{\beta-\alpha}, \quad t>0$$



$$\sim\ y(t) = 0 \longrightarrow t<0$$

$$= \frac{e^{-\alpha t} - e^{-\beta t}}{\beta-\alpha} \longrightarrow t \geq 0$$

Convolution by Slide & Shift Method

Example 1. $x(t) = t^2 + 2t + 1$ and $y(t) = t^2 + 3t + 4$, Find $z(t) = x(t) * y(t)$ by Slide and Shift Method.

$$\Rightarrow z(t) = x(t) * y(t)$$

$$= \int_{-\infty}^{\infty} x(\tau)\, y(t-\tau)\, d\tau$$

$$= \int_{-\infty}^{\infty} x(t-\tau)\, y(\tau)\, d\tau$$

			t^2	$2t$	1	
4	$3t$	t^2				
	4	$3t$	t^2			$= t^4$
		4	$3t$	t^2		$= 3t^3 + 2t^3 = 5t^3$
			4	$3t$	t^2	$= 4t^2 + 6t^2 + t^2 = 11t^2$
				4	$3t$	$= 8t + 3t = 11t$
					4	$= 4$

$$\Rightarrow z(t) = x(t) * y(t)$$

$$= t^4 + 5t^3 + 11t^2 + 11t + 4$$

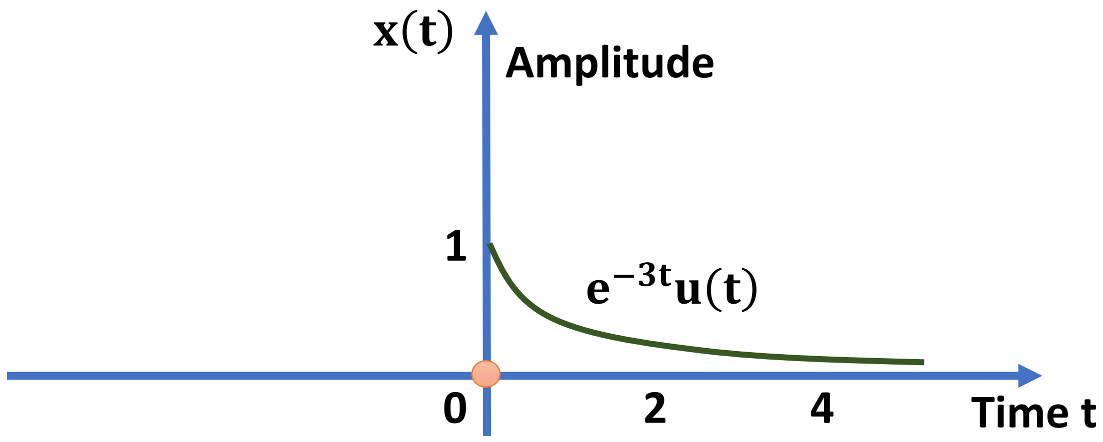
Example 2. $x(t) = 3t^3 + t + 1$ and $y(t) = t^2 + t + 2$, Find $z(t) = x(t) * y(t)$ by Slide and Shift Method.

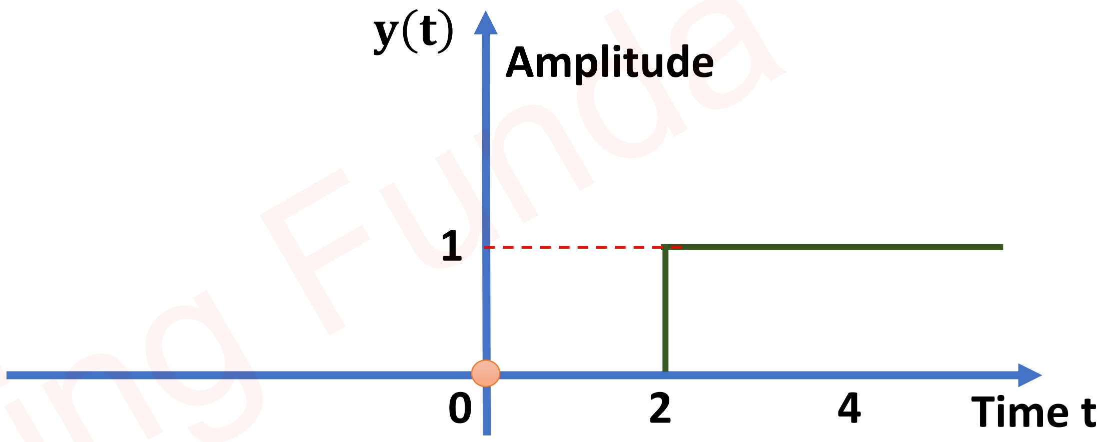
			$3t^3$	0	t	1	
2	t	t^2					
	2	t	t^2				$= 3t^5$
		2	t	t^2			$= 3t^4$
			2	t	t^2		$= 6t^3 + t^3 = 7t^3$
				2	t	t^2	$= t^2 + t^2 = 2t^2$
					2	t	$= 2t + t = 3t$
						2	$= 2$

$$\Rightarrow z(t) = x(t) * y(t)$$
$$= 3t^5 + 3t^4 + 7t^3 + 2t^2 + 3t + 2$$

Convolution by Image Method

Example 1. Find $z(t) = x(t) * y(t)$ by Image Method.





~> Convoluted signal

$$z(t) = x(t) * y(t)$$

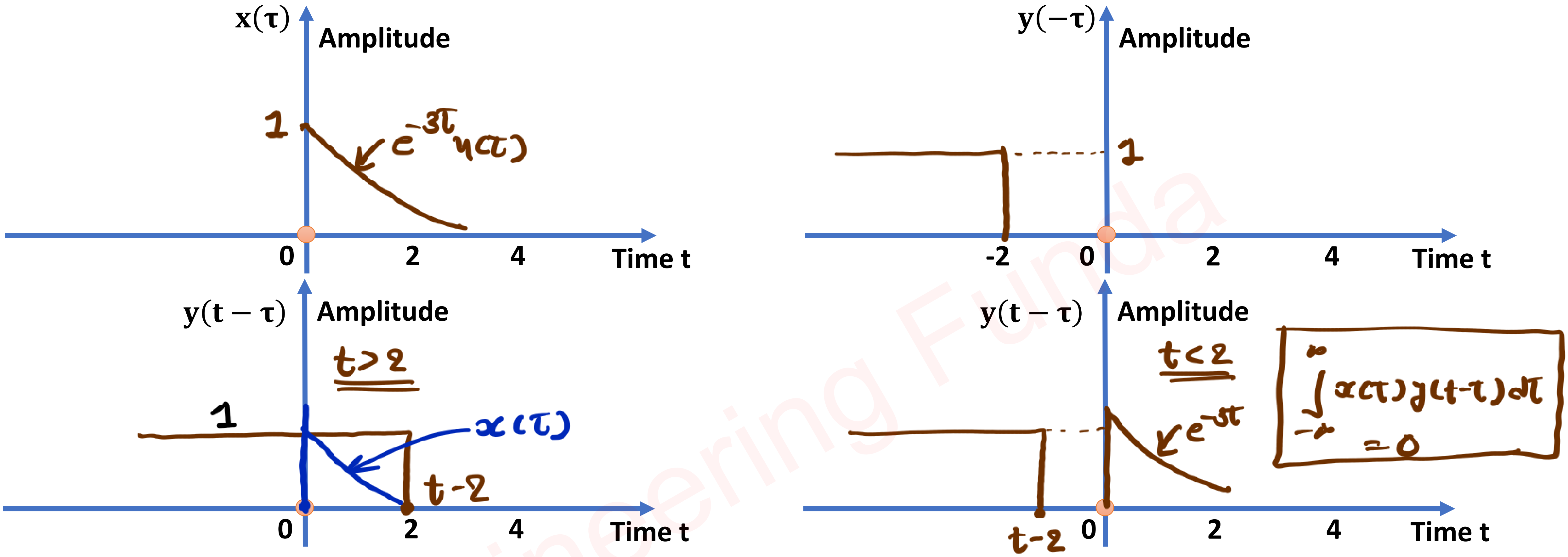
$$= \int_{-\infty}^{\infty} x(\tau)\, y(t-\tau)\, d\tau$$

~> Range of convoluted signal z(t)

$\Rightarrow$ Sum of lower Limits < t < Sum of higher limits.

$\Rightarrow 0+2 < t < \infty + \infty$

$\Rightarrow 2 < t < \infty$



For $t \geq 2$

$$z(t) = \int_0^{t-2} x(\tau)\, y(t-\tau)\, d\tau$$

$$= \int_0^{t-2} e^{-3\tau}\, d\tau$$

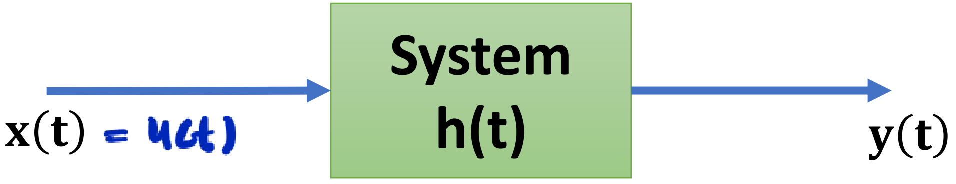
$$\Rightarrow z(t) = \left[\frac{e^{-3\tau}}{-3}\right]_0^{t-2}$$

$$= \frac{1}{3}\left[1 - e^{-3(t-2)}\right], \quad t \geq 2$$

$$= \frac{1}{3}\left[1 - e^{-3(t-2)}\right] u(t-2)$$

Examples of Convolution

Example 1. The Impulse Response of Continuous System is given by: $h(t) = \delta(t-1) + \delta(t-3)$.
The value of Step Response at t = 2 sec is ......1.........



$x(t) = u(t)$

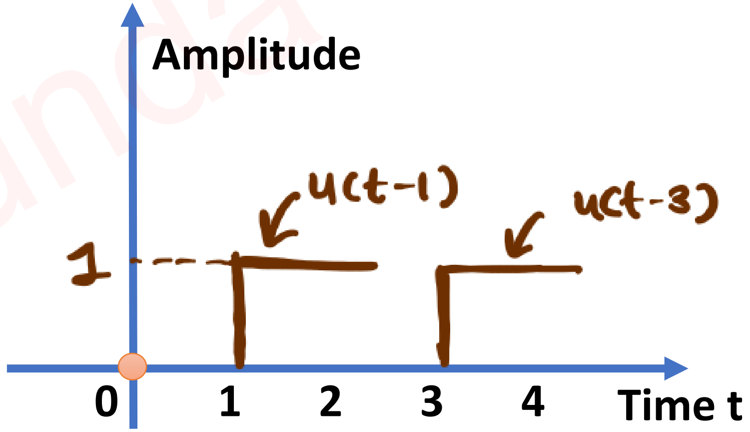
$h(t) = \delta(t-1) + \delta(t-3)$

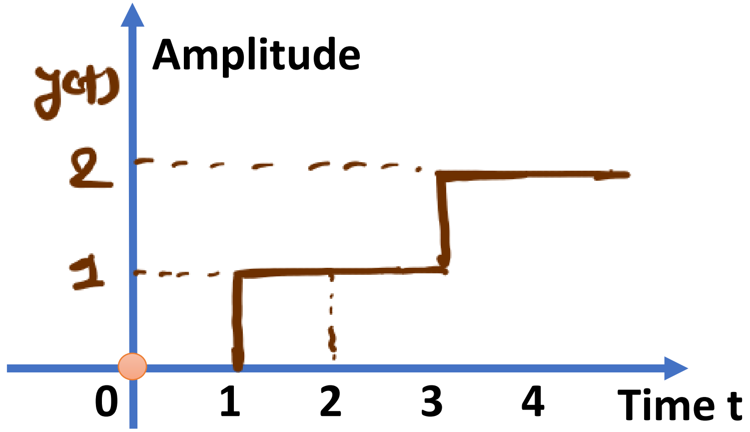
$y(t) = x(t) * h(t)$

$= u(t) * (\delta(t-1) + \delta(t-3))$

$= u(t) * \delta(t-1) + u(t) * \delta(t-3)$

$= u(t-1) + u(t-3)$





Example 2. Given, $\int_{-\infty}^{\infty} x(-\tau+b)y(t+\tau)d\tau$, Express $z(t)$ in terms of $y(t)$, If $y(t)=x(t)*h(t)$.

$\rightsquigarrow y(t) = x(t) * h(t)$

$$= \int_{-\infty}^{\infty} x(\tau)\, h(t-\tau)\, d\tau$$

$$y(t) = \int_{-\infty}^{\infty} x(t-\tau)\, h(\tau)\, d\tau$$

$$\rightsquigarrow z(t) = \int_{-\infty}^{\infty} x(-\tau+b)\, y(t+\tau)\, d\tau$$

$- \; t+\tau = \tau' \Rightarrow \tau = \tau' - t$

$\Rightarrow d\tau = d\tau'$

$$\rightsquigarrow z(t) = \int_{-\infty}^{\infty} x(-(\tau'-t)+b)\, y(\tau')\, d\tau'$$

$$= \int_{-\infty}^{\infty} x((t+b)-\tau')\, y(\tau')\, d\tau'$$

$$= y(t+b)$$

Example 3. If $x(t) = u(t-3) - u(t-5)$, and $h(t) = e^{-3t}u(t)$. Then Find $\frac{dx(t)}{dt} * h(t)$.

$$\Rightarrow \frac{dx(t)}{dt} * h(t) = \frac{d}{dt}\left(u(t-3) - u(t-5)\right) * e^{-3t}u(t)$$

$$= \left(\delta(t-3) - \delta(t-5)\right) * e^{-3t}u(t)$$

$$= \delta(t-3) * e^{-3t}u(t) - \delta(t-5) * e^{-3t}u(t)$$

$$= e^{-3(t-3)}u(t-3) - e^{-3(t-5)}u(t-5)$$

Area of Convoluted Signal

Example 1. Find the area of given convoluted Signal:

$$\big(u(t+1)-u(t-3)\big) * \big(u(t+1)-u(t-1)\big)$$

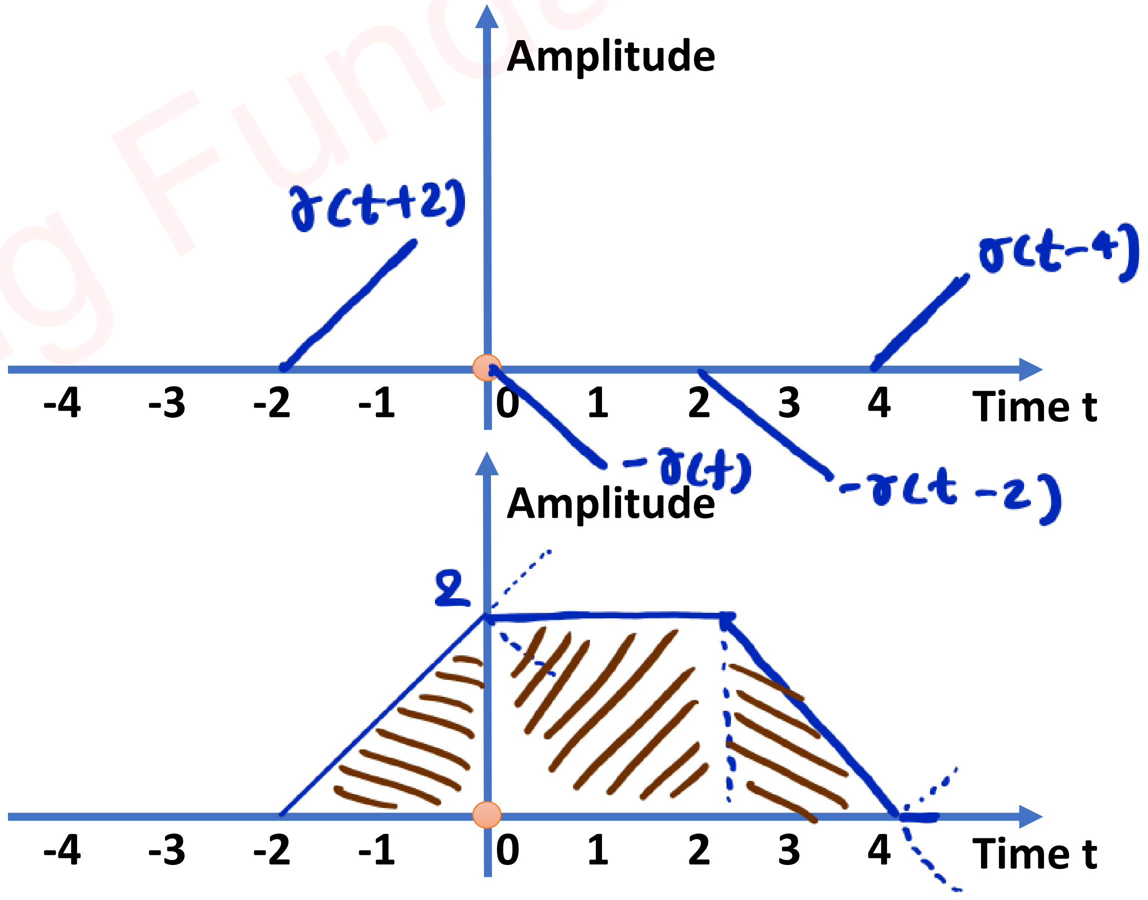
$$= u(t+1) * u(t+1) - u(t+1) * u(t-1) - u(t-3) * u(t+1) + u(t-3) * u(t-1)$$

$$= r(t+2) - r(t) - r(t-2) + r(t-4)$$

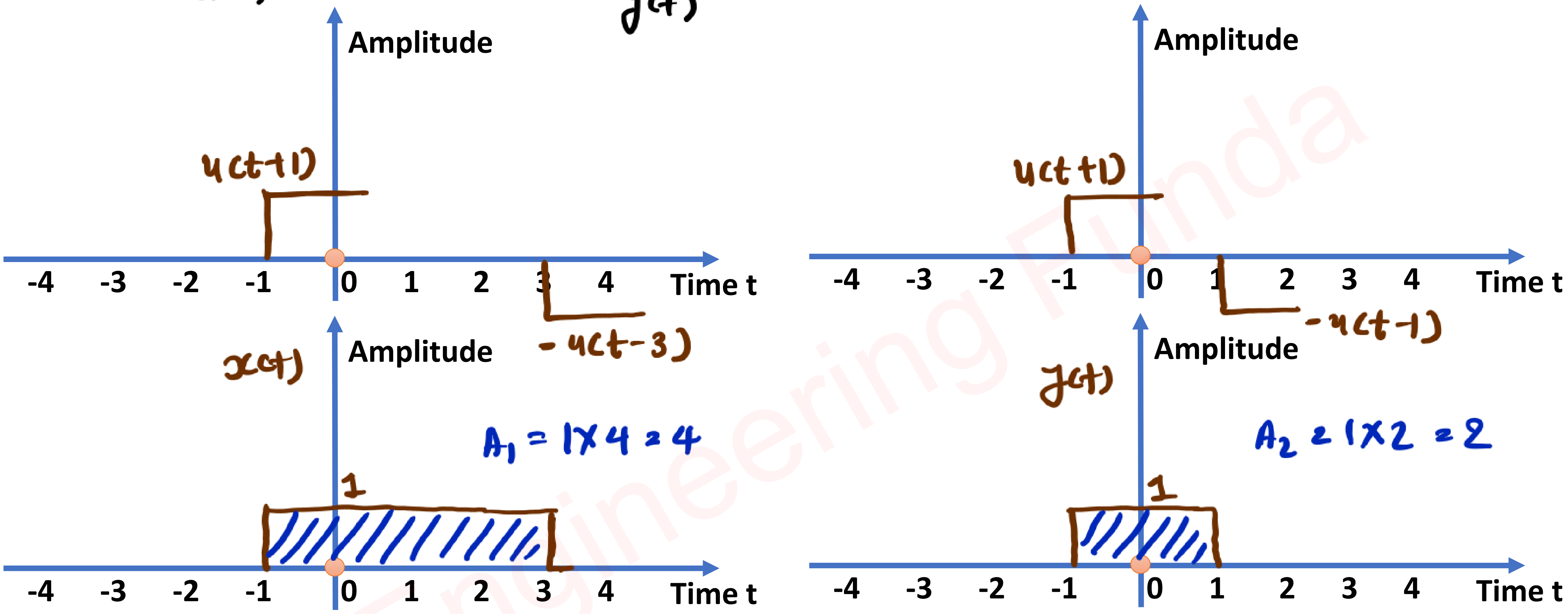
$$\therefore \text{Area} = \frac{1}{2}\times 2\times 2 + 2\times 2 + \frac{1}{2}\times 2\times 2$$

$$= 2+4+2$$

$$= 8$$



$$\rightarrow \Big(\underbrace{u(t+1) - u(t-3)}_{x(t)}\Big) * \Big(\underbrace{u(t+1) - u(t-1)}_{y(t)}\Big)$$



$\rightsquigarrow$ Total area of Convoluted signal $= A_1 A_2$

$$= 4 \times 2$$

$$= 8$$

Linear Convolution

Example 1. $x(n) = \{1, \underset{\uparrow}{2}, 3, 4\}$ and $h(n) = \{1, 1, \underset{\uparrow}{-1}, 1\}$, Find Linear Convolution of above Signals.

→ Total Samples for Convoluted Signal $= m + n - 1$

$= 4 + 4 - 1$

$= 7$

	1	2	3	4
1	1	2	3	4
1	1	2	3	4
-1	-1	-2	-3	-4
1	1	2	3	4

→ Convoluted signal

$y(n) = x(n) * h(n) = \{1, 3, 4, \underset{\uparrow}{6}, 3, -1, 4\}$

Example 2. $x(n) = \{1, 3, \underset{\uparrow}{2}, 1\}$ and $h(n) = \{1, \underset{\uparrow}{2}\}$, **Find Linear Convolution of above Signals.**

→ Total Samples for Convoluted Signal $= m + n - 1$

$= 4 + 2 - 1$

$= 5$

	1	3	2	1
1	1	3	2	1
2	2	6	4	2

→ Convoluted signal

$y(n) = x(n) * h(n)$

$= \{1, 5, 8, \underset{\uparrow}{5}, 2\}$

Example 3. $x(n) = \{1, 2, \underset{\uparrow}{1}, 1, 2\}$ and $h(n) = \{3, \underset{\uparrow}{2}, 3, 1\}$, Find Linear Convolution of above Signals.

→ Total Samples for Convoluted Signal $= m + n - 1$

$= 5 + 4 - 1$

$= 8$

	1	2	1	1	2
3	3	6	3	3	6
2	2	4	2	2	4
3	3	6	3	3	6
1	1	2	1	1	2

→ Convoluted signal

→ $y(n) = x(n) * h(n)$

$= \{3, 8, 10, \underset{\uparrow}{12}, 13, 8, 7, 2\}$

Circular Convolution

Example 1. $x(n) = \{1, 2, 1, 1\}$ and $h(n) = \{1, 1, -1, -1\}$, Find Circular Convolution of above Signals.

↝ Total Samples in linear Convolution = $m+n-1 = 4+4-1 = 7$

	1	2	1	1
1	1	2	1	1
1	1	2	1	1
-1	-1	-2	-1	-1
-1	-1	-2	-1	-1

↝ For linear convolution

$$y_l(n) = \{\underbrace{1, 3, 2, -1}_{-2\ \ -2\ \ -1}, \underbrace{-2, -2, -1}\}$$

↝ Total Samples in Circular Convolution

$= \max(m, n)$

$= 4$

↝ Circular convolution.

$$y_c(n) = \{-1, 1, 1, -1\}$$

Example 2. $x(n) = \{1, 2, 1, 2\}$ **and** $h(n) = \{2, 1, 1\}$**, Find Circular Convolution of above Signals.**

→ Total Samples in linear Convolution $= m+n-1 = 4+3-1 = 6$

	1	2	1	2
2	2	4	2	4
1	1	2	1	2
1	1	2	1	2

→ linear Convolution

$$y_l(n) = \{\underbrace{2, 5, 5, 7}_{3\quad 2}, 3, 2\}$$

→ Total samples in Circular Convolution

$= \text{Max}(m, n)$

$= 4$

→ Circular convolution

$$y_c(n) = \{5, 7, 5, 7\}$$

Circular Convolution by Matrix Method

Example 1. $x(n) = \{1, 1, -1, 2\}$ and $h(n) = \{1, 2, 3, 4\}$, Find Circular Convolution of above Signals.

$$\rightsquigarrow y_c(n) = x(n) \circledast h(n)$$

$$= \begin{bmatrix} 1 & 2 & -1 & 1 \\ 1 & 1 & 2 & -1 \\ -1 & 1 & 1 & 2 \\ 2 & -1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}$$

$$= \begin{bmatrix} 1+4-3+4 \\ 1+2+6-4 \\ -1+2+3+8 \\ 2-2+3+4 \end{bmatrix} = \begin{bmatrix} 6 \\ 5 \\ 12 \\ 7 \end{bmatrix}$$

$$= \{6, 5, 12, 7\}$$

Example 2. $x(n) = \{1, 2, 1, 1\}$ and $h(n) = \{1, 1, -1, -1\}$, **Find Circular Convolution of above Signals.**

$$y_c(n) = x(n) \circledast h(n)$$

$$= \begin{bmatrix} 1 & 1 & 1 & 2 \\ 2 & 1 & 1 & 1 \\ 1 & 2 & 1 & 1 \\ 1 & 1 & 2 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ -1 \\ -1 \end{bmatrix}$$

$$= \begin{bmatrix} 1+1-1-2 \\ 2+1-1-1 \\ 1+2-1-1 \\ 1+1-2-1 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \\ 1 \\ -1 \end{bmatrix}$$

$$= \{-1, 1, 1, -1\}$$

Example 3. $x(n) = \{1, 2, 1, 2\}$ and $h(n) = \{2, 1, 1\}$, **Find Circular Convolution of above Signals.**

$$y_c(n) = x(n) * h(n)$$

$$= \begin{bmatrix} 1 & 2 & 1 & 2 \\ 2 & 1 & 2 & 1 \\ 1 & 2 & 1 & 2 \\ 2 & 1 & 2 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \\ 1 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} 2+2+1 \\ 4+1+2 \\ 2+2+1 \\ 4+1+2 \end{bmatrix} = \begin{bmatrix} 5 \\ 7 \\ 5 \\ 7 \end{bmatrix}$$

$$= \{5, 7, 5, 7\}$$

Deconvolution

Example 1. If a Signal $x(n) = \{1, 1, -1\}$ is convoluted with unknown signal $h(n)$ & Convolution result is noted as $y(n) = \{1, 3, 1, -2\}$, Find $h(n)$.

$\rightsquigarrow$ $y(n) = x(n) * h(n)$

$\rightsquigarrow$ Total Samples $= m+n-1$

$\Rightarrow 4 = 3+n-1$

$\Rightarrow n = 2$

$\rightarrow h(n) = \{a, b\}$

$= \{1, 2\}$

	1	1	-1
a	a	a	-a
b	b	b	-b

$y(n) = \{a, a+b, -a+b, -b\}$

$= \{1, 3, 1, -2\}$

$$\left.\begin{aligned} &- a = 1 \\ &- b = 2 \end{aligned}\right\} \text{Valid Values}$$

- $x(n) = \{1, 1, -1\} = 1 + x - x^2$
- $y(n) = \{1, 3, 1, -2\} = 1 + 3x + x^2 - 2x^3$
- $y(n) = x(n) * h(n) \Rightarrow h(n) = y(n)/x(n)$

$$
\begin{array}{r|l}
 & 1 + 2x \\
\hline
1 + x - x^2 & 1 + 3x + x^2 - 2x^3 \\
 & \underline{-1 \mp x \pm x^2} \\
 & \quad 2x + 2x^2 - 2x^3 \\
 & \quad \underline{-2x \mp 2x^2 \pm 2x^3} \\
 & \qquad 0
\end{array}
$$

Valid

- $h(n) = 1 + 2x$
$= \{1, 2\}$

Example 2. Here, $y(n) = x(n) * h(n)$. If $x(n) = \{1, 1, 2\}$ and $y(n) = \{2, 1, 3, 4, 1\}$, Find $h(n)$.

- $x(n) = \{1, 1, 2\} = 1 + x + 2x^2$
- $y(n) = \{2, 1, 3, 4, 1\} = 2 + x + 3x^2 + 4x^3 + x^4$
- $y(n) = x(n) * h(n) \Rightarrow h(n) = y(n)/x(n)$

$2 - x$ ← h(n) ✗

$1 + x + 2x^2 \,\big)\, 2 + x + 3x^2 + 4x^3 + x^4$

$\underline{\;2 + 2x + 4x^2\;}$ (signs: $-\ -\ -$)

$-x - x^2 + 4x^3$

$\underline{\;-x - x^2 - 2x^3\;}$ (signs: $+\ +\ +$)

$\underline{\;6x^3 + x^4\;}$

↑

It can not be zero

- y(n) is incorrect.